Samuel Garnett

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Education

Rensselaer Polytechnic Institute — **Graduation: May 2027**

Bachelor of Science in Computer Science and Physics
Machine Learning Concentration

- **GPA:** 3.8/4.0
- **Relevant Coursework:** Data Structures, Algorithms, Computer Architecture, Software Design, Linear Algebra
- **Activities:** Quantum Computing Club, RCOS, Sigma Alpha Epsilon, Computer Science Honors Society

Experience

General Dynamics Information Technology — Falls Church, VA
Software Engineering (AI Systems) Intern — *June 2025 – August 2025*

- Implemented advanced LLM reasoning strategies into company-wide GenAI services used by 15000 employees
- Built and deployed full-stack services for UAV flight path prediction, integrating Node.js, Python APIs, and PostgreSQL
- Operationalized backend services across GDIT contracts by building CI/CD pipelines, reducing deployment time by ∼30%

William & Mary University — Williamsburg, VA
Machine Learning Research and Engineering Intern — *August 2024 – January 2025*

- Engineered a speculative decoding algorithm using PyTorch to accelerate token generation in LLMs by over 20%
- Combined LangChain and Pytorch to implement verbal reinforcement learning in 25% smarter multi-LLM reasoning agents

Rensselaer Polytechnic Institute — Troy, NY
Teaching Assistant - Various Courses — *May 2024 – Present*

- Taught and mentored 500+ students across quantum computing, discrete mathematics, and data structures courses
- Graded 300+ exams, created 3 sets of lecture notes, and commited over 2000 lines of code for course websites

Rensselaer Polytechnic Institute — Troy, NY
Research Assistant - Quantum Computing and Machine Learning — *January 2024 – May 2024*

- Created 3 presentations on how reinforcement learning is used to accelerate the classical simulation of quantum circuits
- Presented about the applications of LLMs in the financial sector at the DC summit for AI in Finance

Projects

Operating Systems Process and CPU Scheduling Simulator | ***C++, Linux, Bash/Unix***
Developer

- Built an operating systems simulator modeling process execution, ready queues, CPU scheduling, context switching, and I/O waiting across FCFS, SJF, SRT, and RR algorithms

Multi-LLM Reasoning with Speculative Decoding | ***PyTorch, LangChain, HuggingFace, Bash/Unix***
Co-Founder

- Used speculative decoding to accelerate verbal reinforcement learning in multi-LLM agents by 50% without sacrificing response quality

Multithreaded TCP Wordle Server | ***C++, TCP/IP, Linux, Multithreading***
Developer

- Built a TCP server to host a Wordle-style game, leveraging concurrency for client connections and safely tracking shared global game statistics

Technical Skills

Programming Languages: C++, C, Python, Java, R, Assembly, JavaScript, TypeScript, JSX, TSX, HTML, CSS
Libraries and Frameworks: PyTorch, LangChain, Node.js, React.js, NumPy, Pandas, Spacy, FastAPI
Developer Tools: Git, GitLab, Docker, AWS, Kubernetes, Slurm, Linux, Postman, PostgreSQL, Development Containers
Concepts Full-Stack Development, Machine Learning, DevOps, HPC, Model Finetuning, Git Workflows, CI/CD Pipelines

Awards and Honors

Dean's Honor List: Named on the Dean's Honor List for every semester attending RPI
George A. Strichman Scholar: Named the sole recipient of the George A. Strichman Endowed Scholarship